Challenge — Why current training is hard?

1 Hard questions are expensive to construct



What animal is this?

Cat !

Existing VQA questions often easy…

Constructing a Question…

Collect images → Build relations → Validate evidence → Curate Multimodal data → Human verification

If not pass

Expensive and time-consuming

2 Long-horizon investigations are complex



What prize did the person who hosted an event at the location shown in this image win in his 17th edition?

Reverse Image Search → Text Search → Web Visit → Image Search → Crop Image → Check Location → Explore Related Entities → Visual Inspection

Long horizon and cross-modal

3 Terminal rewards give weak process feedback



… → Redundant Verification

… → Premature stopping

… → Wrong Answer

Need richer feedback and teacher guidance

Representative Case — Same image, harder question after refine.

A Seed Question



Reverse Image Search → Mount Washington Hotel

B Refine

Build Link → Expand → Blur → Visualize

C Refined Question

The grand hotel shown hosted a UN-sponsored international financial meeting attended by delegates from dozens of nations. The monetary system associated with that meeting was ended by a U.S. president's major economic action, and the president's resignation brought in a successor whose presidential library stands on a university campus. In a dusk stadium photograph associated with that university, how many white cylindrical columns are visible beneath the large dark screen?

Long Horizon | More tools | Cross-modal | Candidate pruning | Visual reasoning

D Evidence-seeking Solution

Text Search | Web Visit | Image Search | Crop Image | Reverse Image Search | Fetch Image | Python

1 Stage 1: Identify the building



Input Image

Visual clues extracted:
- Distinctive red roof and white façade
- Large, sprawling resort hotel
- Golf course in foreground
- Mountainous background(white Mts.)



Mount Washington Hotel — Bretton Woods, New Hampshire…



Omni Mount Washington Resort — Bretton Woods, NH Iconic hotel…



The Balsams — Dixville Notch, NH Historic hotel…

Confirmed Entity:



Mount Washington Hotel

2 Stage 2: Bridge the history chain



Bretton Woods Conference — International monetary conference held at Mount Washington Hotel in Bretton Woods, NH…



Attendees — U.S. delegates included Henry Morgenthau Jr)later U.S. Treasury Secretary



Morgenthau's successor — Succeeded by Gerald R.Ford as U.S. Treasury Secretary(1974-1975)…



Gerald Ford — 38th President of the United States(1974-1977)

3 Stage 3: Locate the presidential library

Gerald ford presidential library university

Duke University — Not associated with Gerald Ford's presidential library



Universitas Princetoniensis — Not associated with Gerald Ford's presidential library



University of Michigan — Gerald R. Ford Presidential Library is located in Ann Arbor, MI.

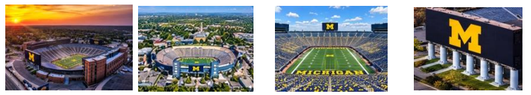


Standford University — Not associated with Gerald Ford's presidential library



4 Stage 4: Find the dusk stadium photo

University of Michigan stadium at dusk



Candidate 1 | Candidate 2 | Candidate 3 | Candidate 4



Selected crop

5 Stage 5: Count the white columns





The answer is 8!!!